Theodor Bergenstof Gilhøj

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Skills

Programming: Python, R, MATLAB, C++, Java, SQL, Gurobi
Software: Git, Simulink, MS Excel, dSPace, ASHES, HAWC2
CAD: PC SCHEMATIC, AutoCAD, NX, SolidWorks, ParaView
Hardware: MicroLabBox, Arduino

Education

DTU – Master of Science in Wind Energy — Expected 2027
DTU – Bachelor of Science in Design of Sustainable Energy Systems, GPA: 10.3 — 2025
KIT – Exchange semester at Karlsruhe Institute of Technology — 2024
Gymnasium / High School – Mathematics, Physics, Chemistry, GPA: 10.6 — 2019
Primary and Secondary School (Folkeskole) – Bagsværd Kostskole, GPA: 10.6 — 2016

Experience

Student worker Systra – Copenhagen, DK — May 2023 – April 2024

- Developed Python scripts to automate repetitive version control and documentation tasks, to cut processing time by orders of magnitude and significantly reduce human errors
- Identified inefficiencies in manual version control processes, and developed Python scripts to automate data extraction and updates across Word documents, to streamline documentation processes and reduce documentation time by >80%
- Produced and updated as-built railway circuit diagrams in AutoCAD, NX, and SolidWorks to deliver accurate system documentation for handover, maintenance, and future modifications
- Documented as-built cable routings, security systems, and relay configurations in PC SCHEMATIC to support safety compliance, troubleshooting, and long-term system reliability

Profile

Elite sport – 1x European Champion and 4x Nordic Champion on the Danish national team in military pentathlon, training 20+ hours a week alongside full-time studies
Music – Lifelong musician with 15+ years of guitar experience and 5+ years of piano, currently taking classical piano lessons
Language – Fluent in Danish, English, and German, with ongoing Spanish studies
Professional motivation

- I enjoy solving complex open-ended problems with a systematic approach, programming, and mathematical modeling
- I am intrinsically driven to deliver high quality results in all aspects of my life, grades are not my main motivator
- Known by peers and supervisors as intellectually curious with an analytical mindset
- Academically strong, and good at communicating technical concepts clearly in writing and conversation

Projects

- **Dynamic shear profile modeling and validation with LiDAR data (Grade: 12) (BSc Project)** — PDF
  - Developed a real-time algorithm in python to estimate wind shear profiles using synthetic LiDAR data produced with HAWC2, that applies advanced filtering and a longitudinal velocity estimation to improve turbine load predictions
  - Evaluated the model with synthetic constant and time-varying shear data, to achieve a relative error of 10% for the shear exponent and 0.7% for the reference wind speed
  - Investigated the impact of estimation frequency and dynamic conditions on model stability, to identify optimization strategies to enhance convergence and robustness for future validation
- **Drive train control of a grid-connected wind turbine (Grade: 10 (Lab Project))** — PDF
  Implemented and lab-tested field-oriented control, active front-end voltage oriented control, and microgrid characterization in Simulink & dSPACE to study torque control, droop behaviour, grid-angle and inertia estimation.
- **Energy Management with Embedded systems Design Project (Grade: 12)** — PDF
  Developed an Arduino-based embedded system to automate dorm-room heating, cooling and lighting using sensors, PWM-controlled actuators, an LCD interface and an IoT dashboard for real-time monitoring, manual override and energy-efficient indoor climate control.
- **Heat Storage Design Project (Grade: 12)** — PDF
  Designed and built a thermally insulated heat storage prototype with electric heating, derived a lumped thermal model and Python simulation to compare theoretical cooldown with measured temperature data over multiple phases.

- **Transformer Virtual Design Project (Grade: 100%)** PDF
  Developed a Python-based design workflow for a distribution transformer by scripting the full electromagnetic and thermal design, iterating geometry and materials under efficiency, short-circuit and voltage-regulation constraints
- **Wind Turbine Rotor Design Project (Grade: 12)** PDF
  Designed and built a two-bladed wind turbine rotor based on airfoil selection, Python-based blade element theory for chord and twist distribution, and wind-tunnel testing to compare thrust, torque and power coefficient with theory
- **Flywheel Design Project (Grade: 12)** PDF
  Developed and modelled a flywheel energy storage concept for city buses, combining prototype design, lab testing and Python-based dynamic simulations to size materials and geometry, estimate fuel savings and refine the business case
- **Projection of thermal images onto 3D turbine blade surfaces (KIT lab project) (Grade: A)** PDF
  Projected infrared images from turbine blade tests onto 3D CAD geometry using calibrated camera models and Python tools, and validated cooling channel alignment in ParaView
- **Electricity market modelling with renewables (Grade: 12)** PDF
  Implemented multi-period DC optimal power flow and storage models in Python for a 24-bus system, structured large datasets and generated visual analyses of congestion, prices, and storage effects, to support decision making for TSO's.
- **Case Project: Power Plant Operational Analysis (Innovation project, Grade: 12)** PDF
  Analysed a local district heating plant's economics, heat-plant sizing (capacity, volume flow, etc.) and time series of heat delivery to the city, to propose a PV–battery solution combined with an educational science park and a facade restoration

Additional Professional Experience

Tutor DinTutor / Privat – Copenhagen, DK 2018 – 2022

- Raised Mathematics and Physics performance for gymnasium students by assessing knowledge gaps and delivering targeted exam preparation, resulting in grade improvements from the bottom to the very top of the grade scale

Professional Soldier Danish Defence – Slagelse, DK 2019 – 2021

- Executed tasks under strict procedural and quality requirements, worked closely in a team, with responsibility at both individual and unit-level, to deliver accurate results under time pressure and rough conditions

Labor Worker Freelance – Brisbane, Australia Aug 2019 – Jan 2020

- Performed physically demanding hands-on work across multiple trades, adapted quickly to new processes and tools, and ensured quality under tight deadlines, both on duty alone and in cooperation with others

Youth Coach Ballerup Baseball Softball Club – Ballerup , DK 2015 – 2018

- Organized and led training sessions twice a week, which developed leadership, planning, and team coordination skills